To the Editors
Discover Artificial Intelligence

Re: Article submission — "Same Transactions, Different Alarms: Foundation-Model Choice and AML False Positives."

Dear Editors,

We are pleased to submit the above manuscript for consideration as a Research article in *Discover Artificial Intelligence*. Using a pre-registered design, we measure how much the choice of foundation model changes the operating point of large-language-model anti-money-laundering (AML) screening when the task and transaction cases are held fixed. Five commercially available models from two providers were run as zero-shot screeners over an identical 600-case battery anchored to FATF laundering typologies and established synthetic-AML data structures (12,000 elicitations across two prompt phrasings and two seeds). On the same 300 legitimate cases, false-positive rates ranged from 0.3% to 83.0% — an 82.7 percentage-point spread (Cochran's $Q = 131.6$, $P \approx 2 \times 10^{-27}$) — the models disagreed on 85.7% of legitimate cases, and within each provider the rate fell monotonically with model tier. Projected to an illustrative low-prevalence scenario, this choice changes analyst alert workload about 200-fold on identical activity. Our estimand throughout is the between-model change in operating point, not accuracy.

The work fits *Discover Artificial Intelligence* through its relevance to the reliability and governance of AI decision systems in finance and its methodological rigor: the design is pre-registered, inference is paired and distribution-free (Cochran's Q, McNemar's exact test with Bonferroni correction, Wilson score intervals, and a cluster bootstrap), non-LLM rules and supervised baselines provide a reference frame, and a deterministic, offline reproducibility capsule (Code Ocean DOI 10.24433/CO.4804007.v1; MIT license) regenerates every reported number, table, and figure byte-for-byte. The study uses only synthetic transaction data, involves no human or animal subjects, and discloses its use of generative-AI assistance in the Declarations.

The manuscript is original, is not under consideration elsewhere, and has been approved by both authors, who declare no competing interests and received no external funding. We have not discussed this work with a *Discover Artificial Intelligence* Editorial Board Member and request no referee exclusions. Thank you for your consideration.

Sincerely,

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